

Miscellaneous errata and style changes for:
Mathematical Control Theory: Deterministic Finite Dimensional Systems
(Eduardo D. Sontag, Springer, New York, 1990).

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With special thanks to: Sarah Koskie, Gerardo Lafferriere, Konrad Reif.

Legend: “[c]” indicates change must be made; “[o]” indicates optional style change.

- [c] **page 002**, Figure 1.1: arrow labeled $-mg \sin \theta$ should be smaller than vertical arrow.
 - [o] **page 026**, line 10: “ $\omega \diamond = \diamond \omega$ ” should be replaced by “ $\omega \diamond = \diamond \omega = \omega$ ”.
 - [o] **page 032**, line -11: “ $(\tau + \mu, \sigma + \mu, \nu)$ is again in $\mathcal{D}_\lambda$, where ν ” should be replaced by “ $(\tau + \mu, \sigma + \mu, \omega^\mu)$ is again in $\mathcal{D}_\lambda$, where ω^μ ”.
 - [c] **page 033**, line 15 (display): “ $\mathcal{E} \subseteq \mathbb{Z} \times \mathcal{X} \times \mathcal{U}$.” should be replaced by “ $\mathcal{E} = \mathcal{E}(\phi) = \{(t, x, u) \in \mathbb{Z} \times \mathcal{X} \times \mathcal{U} \mid (t + 1, t, x, u) \in D_\phi\}$.”.
 - [c] **page 033**, line 16: “In the particular case of complete systems, this is an equality.” should be replaced by “In the particular case of complete systems, $\mathcal{E} = \mathbb{Z} \times \mathcal{X} \times \mathcal{U}$.”.
 - [c] **page 033**, line -8: “undefined at $(t, t + 2, x, \omega)$ otherwise” should be replaced by “undefined at $(t + 2, t, x, \omega)$ otherwise”.
 - [c] **page 033**, line -4: “system Σ such that” should be replaced by “system $\Sigma = (\mathbb{Z}, \mathcal{X}, \mathcal{U}, \phi)$ such that $\mathcal{E} = \mathcal{E}(\phi)$ and”.
 - [o] **page 034**, line 11: “For systems with outputs and $\mathcal{Y}$ ” should be replaced by “For systems with outputs, and $\mathcal{Y}$ ”.
 - [o] **page 034**, line -1: Delete equation number.
 - [o] **page 035**, line 1/5: Replace these lines with the following text: “We formalize the problem as that of finding an initialized discrete-time system (Σ, x^0) with outputs whose i/o behavior matches λ for $\tau \geq 1$, that is,
- $$\lambda_{\Sigma, x^0}^{0, \tau} = \lambda^{0, \tau} \quad (2.11)$$
- for all $\tau \geq 1$. Here we will just guess a solution. One possibility is as follows: Consider the complete system with”
- [o] **page 039**, line 7/8 (L/E 2.4.7): “The linear i/o behavior Λ is time-invariant if and only if there exists a sequence of linear maps $\{\mathcal{A}_i, i = 1, 2, 3, \dots\}$ ” should be replaced by “Let Λ be a linear i/o behavior. Consider the family of linear maps $\{\tilde{\mathcal{A}}_{ij}\}_{i > j}$ introduced in Lemma 2.4.5. Then, Λ is time-invariant if and only if there exists a sequence of linear maps $\{\mathcal{A}_i, i = 1, 2, 3, \dots\}$ ”.
 - [o] **page 045**, line 9 (L/E2.6.3): Drop “, by definition” (was meant to mean “by definition of admissibility”, but some readers find this confusing).
 - [c] **page 046**, line 14 (L2.6.8): “denote the dynamics map of” should be replaced by “denote the transition map of”.
 - [c] **page 046**, line -7 “and $\lambda^{\sigma, \tau}$ is continuous .” should be replaced by “and $\lambda^{\sigma, \tau}$ is continuous.”. (move period closer).
 - [o] **page 047**, line 3 (R2.6.11): “It is also true that the complete i/o map ” should be replaced by “It is also true that the complete i/o map $\omega \mapsto \bar{\lambda}_\Sigma(\tau, \sigma, \omega)$ ”.
 - [o] **page 047**, line 9: after “In this section, $\mathcal{K} = \mathbb{R}$ or $\mathbb{C}$.” add: “Appendix C.5 on linear differential equations summarizes needed background material.”

- [o] **page 050**, line -8 (E2.7.12): “fixed $i = 1, \dots, m$ and any” should be replaced by “fixed $i \in \{1, \dots, m\}$ and any”.
- [c] **page 051**, line 20 (D2.7.14): “local-in-time description $(\mathbb{R}^n, \mathbb{R}^m, f_*)$, where” should be replaced by “local-in-time description f_* , where”.
- [o] **page 052**, line 3/4: “if (x, u) is like this” should be replaced by “if (x, u) is an equilibrium pair”.
- [c] **page 052**, line after (2.31): “the the” should be replaced by “the”.
- [c] **page 054**, (equation (2.33)): “ $\alpha_*[0, (0, 0)] : \mathbb{R}^{n+m^2} \rightarrow \mathbb{R}^n$:” should be replaced by “ $\alpha_*[0, (0, 0)] : \mathbb{R}^{n+2m} \rightarrow \mathbb{R}^n$:”.
- [o] **page 055**, line -5: “for every integrable function from $[\sigma, \tau]$ to $\mathcal{K}^m$.” should be replaced by “for every integrable function $\varphi : [\sigma, \tau] \rightarrow \mathcal{K}^m$.”. Also add: “(This is the weak-* topology on $\mathcal{L}^\infty$ seen as dual of $\mathcal{L}^1$.)”
- [c] **page 056/57**, (Theorem 1): *There are several changes here. The new version of these two pages can be found in a separate file.*
- [o] **page 058**, line 7: “because N has finite support” should be replaced by “because N has compact support”.
- [o] **page 058**, line 14: “For any other control ν (of length T)” should be replaced by “For any other control $\nu \in \mathcal{L}_m^\infty(\sigma, \tau)$ ”.
- [o] **page 058**, line -10: “From Equations (2.36) and (2.38) we conclude that” should be replaced by “From (2.36) and (2.38) we conclude that”.
- [c] **page 061**, line -3 (L/E2.8.3): “ $D\alpha(x, u)[\lambda_0, \mu]$ ” should be replaced by “ $\alpha_*[x, \omega](\lambda_0, \mu)$ ”. Also, in lines (-5, -1) “ u ” should be replaced by “ ω ”. (in 4 places).
- [o] **page 062**, line 10: “When $\mathcal{U} \subseteq \mathbb{R}^m$, we may consider for each $p \geq 1$ ” should be replaced by “When $\mathcal{U} \subseteq \mathbb{R}^m$, we may consider for each $p \geq 1$ and each $\sigma < \tau$ ”.
- [o] **page 062**, line -6: “ $c_1 = T^{\frac{1}{p} - \frac{1}{q}}$ ” should be replaced by “ $c_1 = (\tau - \sigma)^{\frac{1}{p} - \frac{1}{q}}$ ”.
- [c] **page 065**, line 2: Drop “, with Σ now of linear growth on controls.”
- [c] **page 065**, line 3: Start sentence with: “If S has linear growth on controls, then” and make “The” lower case.
- [c] **page 066**, line 19: “ (x, ω) with $x = \xi(t)$,” should be replaced by “ $(\xi(t), \omega(t))$,”.
- [c] **page 066**, line 22: “Continuity of α ” should be replaced by “Continuity of ψ and α ”.
- [c] **page 070**, line -6 (Exercise 2.10.2): “the following Σ and δ ” should be replaced by “the following Σ and $\delta = 1$ ”.
- [o] **page 072**, line 4: “are matrices of locally bounded functions” should be replaced by “are matrices of locally essentially bounded functions”.
- [c] **page 077**, line 4: “There is a rich an extensive” should be replaced by “There is a rich and extensive”.
- [c] **page 081**, line -5: “where P is the matrix” should be replaced by “where P is the linear transformation”.
- [o] **page 083**, line 18: “It is of interest to know if reachability” should be replaced by “It is of interest to know whether reachability”.

- [o] **page 085**, line 11: “If it would be true that $\mathcal{X}_i$ is properly included” should be replaced by “If it were true that $\mathcal{X}_i$ were properly included”.
- [o] **page 087**, line 17 (after (3.4)): “which means that there exists a linear function” should be replaced by “which means that there exists a linear function (namely, $x \mapsto \rho x$)”.
- [o] **page 089**, line -1: “coefficient of friction on the cart” should be replaced by “coefficient of friction associated with the motion of the cart”.
- [o] **page 091**, line 1: “this is then the same as the set” should be replaced by “ $\mathcal{R}$ is then the same as the set”.
- [o] **page 092**, line 8: “cannot be controlled in any way, and an analogous” should be replaced by “cannot be controlled in any way. An analogous”.
- [o] **page 093**, line 13: “Assume that the rank would be” should be replaced by “Assume that the rank is”.
- [o] **page 093**, line 14: “for some $\lambda \in \tilde{\mathcal{K}}$ ” should be replaced by “for some $\lambda \in \mathcal{K}$ ”.
- [c] **page 093**, line -4/-3: “ $[(\lambda I - A)T^{-1}, B]$ ” should be replaced by “ $[(\lambda I - A)T, B]$ ”.
- [c] **page 098**, line 21 (statement of Lemma 3.4.1): “integer k ” should be replaced by “integer k ”.
- [o] **page 098**, line 23: “only necessary to prove that $A^{(\delta)}$ is invertible.” should be replaced by “only necessary to prove that $A^{(\delta)}$ is invertible if and only if the eigenvalue condition holds.”.
- [o] **page 098**, line -6: “the numerator being nonzero by hypothesis.” should be replaced by “the numerator being nonzero for all $\lambda \neq 0$ if and only if the eigenvalue condition holds.”.
- [c] **page 099**, line 18: “which according to the above Theorem” should be replaced by “which according to the above Lemma”.
- [o] **page 101**, line 8: “Note that the frequencies are defined independently of B .” should be replaced by “Note that B does not affect the frequencies.”.
- [o] **page 102**, line -10: “non-necessarily time-invariant linear systems.” should be replaced by “linear systems, possibly time-varying.”.
- [o] **page 103**, line 21: “It is a Hilbert space with the inner product” should be replaced by “The set $\mathcal{L}_m^2(\sigma, \tau)$ is a Hilbert space with the inner product”.
- [o] **page 103**, line -7: “ $\langle L\omega, x \rangle = \langle \omega, L^*x \rangle$ ” should be replaced by “ $\langle L\omega, x \rangle_{\mathcal{X}} = \langle \omega, L^*x \rangle_{\Omega}$ ”.
- [c] **page 111**, line 16: “ $\text{rank } C\mathcal{R}(A, B) =$ ” should be replaced by “ $\text{rank } C\mathbf{R}(A, B) =$ ”.
- [o] **page 111**, line -4: “smoothly varying on $\mathcal{I}$, and a nontrivial” should be replaced by “smoothly varying on $\mathcal{I}$, and fix a nontrivial”.
- [o] **page 112**, line 15: “If Σ would not be controllable on $[\sigma, \tau]$,” should be replaced by “If Σ were not controllable on $[\sigma, \tau]$,”.
- [c] **page 114**, line 6: “Example 3.5.19” should be replaced by “Exercise 3.5.19”.
- [o] **page 117**, line 8: “there is always because of uniform continuity” should be replaced by “there is always, because of uniform continuity of ϕ in x ,”.
- [o] **page 121**, line -3: “say with state space $\mathcal{Q}$ = open subset” should be replaced by “say with state space $\mathcal{Q}$ an open subset”.
- [o] **page 122**, line 6: After “again of class $\mathcal{C}^\infty$.”, insert sentence: “(We think of $\dot{q} = Q(q)$, $u = h(q)$ as an external system generating controls.)”

[o] **page 122**, line 20/22: “ $\mathbb{R}^k$ ” should be replaced by “ $\mathbb{R}^\ell$ ”. (3 times)

[o] **page 122**, line 25/28: “Whenever defined, one also denotes

$$\mathrm{ad}_f^k(g) := [f, [f, \dots, [f, [f, g]] \dots]]$$

for integers $k \geq 1$ and also $\mathrm{ad}_f^0(g) := g$.” should be replaced by “One also denotes, for each fixed f , $\mathrm{ad}_f(g) := [f, g]$, seen as an operator on vector fields g , and ad_f^k for its k th iterate, if k is a positive integer and the corresponding derivatives exist. Thus,

$$\mathrm{ad}_f^k(g) := \underbrace{[f, [f, \dots, [f, [f, g]] \dots]]}_{k \text{ times}}.$$

[o] **page 123**, line 1/2: “where f_{u_i} denotes the partial derivative of f with respect to the i th coordinate of $\mathcal{U}$ ” should be replaced by “where $f_{u_i}[x, h(q)]$ denotes the partial derivative of f with respect to the i th coordinate of $\mathcal{U}$, evaluated at $(x, h(q))$ ”.

[o] **page 123**, line -8: “It is easy to see that there is an equation

$$(q+2)G + \frac{1}{2}\mathrm{ad}_F^2 G = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

so the desired rank is indeed $1 = \dim \mathcal{X}$ for all q, x and the” should be replaced by “It can be verified that

$$(q+2)G + \frac{1}{2}\mathrm{ad}_F^2 G = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

so the rank in Lemma 3.6.13 is at least one for all q, x . The rank cannot exceed $\dim \mathcal{X} = 1$. Thus, the”.

[o] **page 125**, line -6: Delete “some $\sigma \leq \tau$ and”.

[o] **page 127**, line 30: “that this is simply the property that a certain group action” should be replaced by “that it is simply the property that a certain group action”.

[o] **page 128**, line 3: “is only asked to be dense” should be replaced by “is required only to be dense”.

[o] **page 132**, line 8: “ $\chi(z) = \det(zI - A) = z^n - \alpha_n z^{n-1} - \dots - \alpha_2 z - \alpha_1$ ” should be replaced by “ $\chi(s) = \det(sI - A) = s^n - \alpha_n s^{n-1} - \dots - \alpha_2 s - \alpha_1$ ”.

[o] **page 135**, line -2: Insert before this line “This implies, in particular, that $Ax_k \in \mathcal{V}$. Thus,” and delete the sentence “Furthermore, also by (4.2) —applied in the particular case $u = 0$ — it follows that $Ax_k \in \mathcal{V}$.” in the next page.

[c] **page 140**, line 11: “orbits under this action” should be replaced by “equivalence classes”.

[c] **page 142**, line -3 and -1: There should be no tildes over κ ’s (four places).

[c] **page 144**, line 13: “ $Ae_{i\kappa_i} = -\sum_{j=1}^r \alpha_{ij\kappa_i} b_j$,” should be replaced by “ $Ae_{i\kappa_i} = -\sum_{j=1}^r \alpha_{ij\kappa_i} b_j$,”.

[c] **page 152**, line -1: “ $x := T^{-1}\begin{pmatrix} 0 \\ z_2 \end{pmatrix}$ ” should be replaced by “ $x := T\begin{pmatrix} 0 \\ z_2 \end{pmatrix}$ ”.

[c] **page 153**, line 3: “also $\zeta(t) := T\xi(t)$ converges” should be replaced by “also $\zeta(t) := T^{-1}\xi(t)$ converges”.

[o] **page 158**, line -12 (display): “ $V(\xi(t)) \leq V(x)$ for all $t \in (0, \sigma]$ ” should be replaced by “ $V(\xi(t)) \leq V(x)$ for all $t \in [0, \sigma]$ ”.

[o] **page 158**, line -9: “**relative to** x^0 ” should be replaced by “relative to x^0 ”. (no boldface, so consistent with local definition).

- [c] **page 161**, line 17: “where C is an $p \times n$ matrix” should be replaced by “where C is a $p \times n$ matrix”.
- [c] **page 161**, line 20: “State and prove an analogue of Lemma” should be replaced by “State and prove an analogue of Exercise”.
- [c] **page 167**, line 11 (Ex. 4.6.13): “Consider any system with $\mathcal{X} = \mathcal{R}$, $\mathcal{U} = \mathcal{R}$ ” should be replaced by “Consider any system with $\mathcal{X} = \mathbb{R}^2$, $\mathcal{U} = \mathbb{R}$ ”.
- [c] **page 169**, line -6: Delete “(in fact, A must have an eigenvalue with zero real part)”
- [c] **page 173**, line 11/12: “ $\dot{\xi}_1$ ” should be replaced by “ $\dot{x}_1$ ”. and “ $\dot{\xi}_2$ ” should be replaced by “ $\dot{x}_2$ ”.
- [c] **page 178**, line -9: “the maximum value” should be replaced by “an upper bound on the values”.
- [o] **page 188**, line -12: “The proof of this Lemma was given independently by” should be replaced by “The proof of this Lemma was given independently in”.
- [c] **page 190**, line 1: “ (τ, σ, ω) with $\sigma \leq \tau$ admissible for x^0 ” should be replaced by “ (τ, σ, ω) with ω admissible for x^0 ”.
- [o] **page 195**, line 4,6,7: Drop words “distinct” and “different”. (Note that any state is finite state distinguishable from itself. The terminology is not very good. Better to use “finite state determines if starting from”.)
- [o] **page 195**, line -2 (Ex. 5.1.10): “Show that there exists some $T > 0$ and some fixed control” should be replaced by “Show that there exists some $T \geq 0$ and some fixed control”.
- [c] **page 200**, line -5: “where A_1 is $r \times r$, C_1 is $r \times m$ ” should be replaced by “where A_1 is $r \times r$, C_1 is $p \times r$ ”.
- [c] **page 210**, line -11: “if the intial state is x .” should be replaced by “if the initial state is x ”.
- [o] **page 225**, line 17: After “This establishes properness.” add: “(In fact, we proved that the sequence T_i itself is convergent, rather than merely a subsequence.)”
- [o] **page 227**, line 5: “(which is also written as $W = q^{-1}P$).” should be replaced by “(which is also written as $W = q^{-1}P$, or P/q).”.
- [o] **page 245**, line -9/-7 “For linear (finite dimensional) systems over $\mathcal{K} = \mathbb{R}$ or $\mathbb{C}$, we always take $x^0 = 0$ $u^0 = 0$ $y^0 = 0$ in the definition of asymptotic observability.” should be replaced by “Let Σ be a linear (finite dimensional) system over $\mathcal{K} = \mathbb{R}$ or $\mathbb{C}$. We say simply that Σ is asymptotically observable (or equivalently, detectable) if it is so with respect to $x^0 = 0$ $u^0 = 0$ $y^0 = 0$.”.
- [c] **page 248**, line -4: “However, an equation for $z := q - Lx_1$ can” should be replaced by “However, an equation for $z := q + Lx_1$ can”.
- [c] **page 252**, line 12/13: “1. There is a dynamic stabilizer for Σ_1 .” should be replaced by “1. There is a dynamic stabilizer for Σ_1 (with respect to $x^0 = 0$).”. “2. There is a dynamic linear stabilizer for Σ_1 .” should be replaced by “2. There is a dynamic linear stabilizer for Σ_1 (with respect to $x^0 = 0$).”.
- [c] **page 252**, line -10: “must be asymptotically controllable” should be replaced by “must be asymptotically controllable”.
- [c] **page 253**, insert paragraph after first paragraph (between lines 3 and 4): “We have proved that the system Σ is detectable with respect to x_1^0, u_1^0, y_1^0 . When, as in the statement of the Theorem, Σ_1 is a linear system and $x^0 = 0$, and using that $y_1^0 = k_1(0) = 0$, we have then that the following property holds (using for definiteness the continuous-time case, but the discrete-time argument is exactly the same): for every state trajectory $\xi(t)$ so that $\dot{\xi} = A\xi + Bu_1^0$ and $C\xi(t) \equiv 0$, necessarily

$\xi(t) \rightarrow 0$ as $t \rightarrow \infty$. On the other hand, $(0, u_1^0)$ is an equilibrium pair, so $0 \equiv A0 + Bu_1^0$, which implies that $Bu_1^0 = 0$. Thus it is also true that for every state trajectory $\xi(t)$ so that $\dot{\xi} = A\xi$ and $C\xi(t) \equiv 0$, necessarily $\xi(t) \rightarrow 0$ as $t \rightarrow \infty$, and this means that the system is detectable with respect to $0, 0, 0$, which is what we mean when we say that the linear system Σ_1 is asymptotically controllable.”

- [c] **page 253**, line -3: “ $Dz + Ex$ ” should be replaced by “ $Dz + Eh(x)$ ”.
- [o] **page 259**, line -5: “measurable matrix function $F(t)$ of exponential” should be replaced by “measurable matrix function F defined on $[0, \infty)$ and of exponential”.
- [o] **page 261**, line 1: “as $t \rightarrow \infty$. Thus,” should be replaced by “as $t \rightarrow \infty$. Therefore,”.
- [c] **page 276**, line -1: “ $N := \text{card}(X)$, $M := \text{card}(U)$ ” should be replaced by “ $N := \text{card}(\mathcal{X})$, $M := \text{card}(\mathcal{U})$ ”.
- [o] **page 325**, line 14: “and consider the entire function $h := f.g$ ” should be replaced by “and consider the entire function $h := f \cdot g$ ”.
- [c] **page 341**, line 9: “measurable sets $J_1, \dots, J_k$ ” should be replaced by “measurable sets $J_0, \dots, J_k$ ”.
- [c] **page 343**, line 12: “ $\xi(t) = \int_{\sigma}^t h(\tau) d\tau$ ” should be replaced by “ $\xi(t) = \xi(\sigma) + \int_{\sigma}^t h(\tau) d\tau$ ”.
- [c] **page 344**, line 1: “as a complete metric space when endowed” should be replaced by “as a metric space (itself complete, if $\mathcal{U}$ is complete) when endowed”.
- [c] **page 344**, line 4: No indentation in this line.
- [c] **page 346**, line 3: No space before period after word “Lemma”.
- [o] **page 348**, line 10: “the assumptions imply also a stronger version of Property” should be replaced by “the assumptions imply also a stronger version of Condition”.
- [c] **page 351**, line -4: “ $\mathcal{K}$ ” should be replaced by “ K ”.
- [o] **page 350**, line 10: “Choose again ρ as in Property” should be replaced by “Choose again ρ as in Condition”.
- [c] **page 357-359**, (Appendix C.4): *There are several changes here. The new version of Appendix C.4 can be found in a separate file.*
- [c] **page 361**, line 21: “ $A = Te^{\Lambda}T^{-1}$ ” should be replaced by “ $e^A = Te^{\Lambda}T^{-1}$ ”.
- [c] **page 378**, line 8/9: “1974, pp. 643-638.” should be replaced by “1974, pp. 636-638.”.